Quick Start Guide

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1. First Boot

Clean System Image: Suitable for developers, requires some development experience, and the environment setup process needs independent problem-solving skills!

Booting the Factory Image: Suitable for beginners, allowing you to experience various examples without setting up your own environment! Since the official Jetson AGX Orin's internal 64GB emMC already comes with an outdated system, it needs to be reconfigured to Jetpack 6.2.1 to use the SSD factory image. Please refer to the tutorial [**Chapter 2 Jetson basic course: 02. 02.Flash new system to the onboard eMMC**] to burn the Jetpack 6.2.1 system version, power off the device, insert the factory SSD, and power on again to boot from the SSD.

1.1 Instructions for Use

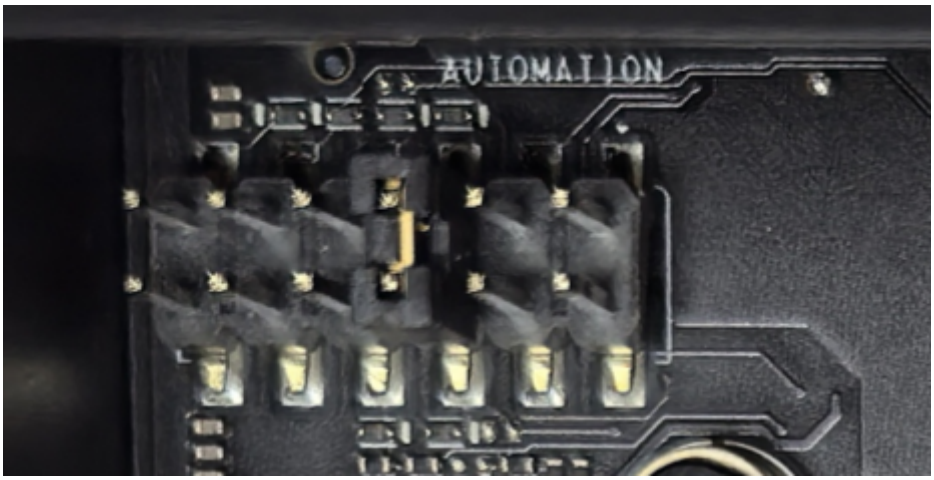
1. SSD Booting: The Jetson AGX Orin's SSD has higher boot priority than the onboard EMMC. Therefore, as long as the SSD and EMMC have the same system version, plugging in the SSD will boot from it first, and unplugging the SSD will boot from the eMMC. Please refer to the tutorial [**Chapter 2 Jetson basic course: 02.Flash new system to the onboard eMMC**] to first burn the system version corresponding to the factory SSD, then power off the device, insert the factory SSD, and power on again to boot from the SSD.
2. SSD Expansion: See Tutorial [**Chapter 2 Motherboard Basics: SSD Expansion**]
3. Burning a Clean Jetson System to the Onboard EMMC: See Tutorial [**Chapter 2 Jetson basic course: 02.Flash new system to the onboard eMMC**]
4. Burning a Clean Jetson System to an NVMe SSD: See Tutorial [**Chapter 2 Motherboard Basics: 03. Flash new system to an NVMe SSD**]
5. Restoring the Factory Image System: See Tutorial [**Chapter 2 Motherboard Basics:Restore initial image system**]
6. Installing the Jetson Component Environment: See Tutorial [**Chapter 2 Motherboard Basics: Installing the Jetpack Environment**]
7. The Jetson AGX Orin motherboard supports burning a clean Jetson system using SDK Manager, but does not have a SUPER mode.
8. The Jetson AGX Orin motherboard comes with a built-in wireless network card and antenna.

1.2 System Booting

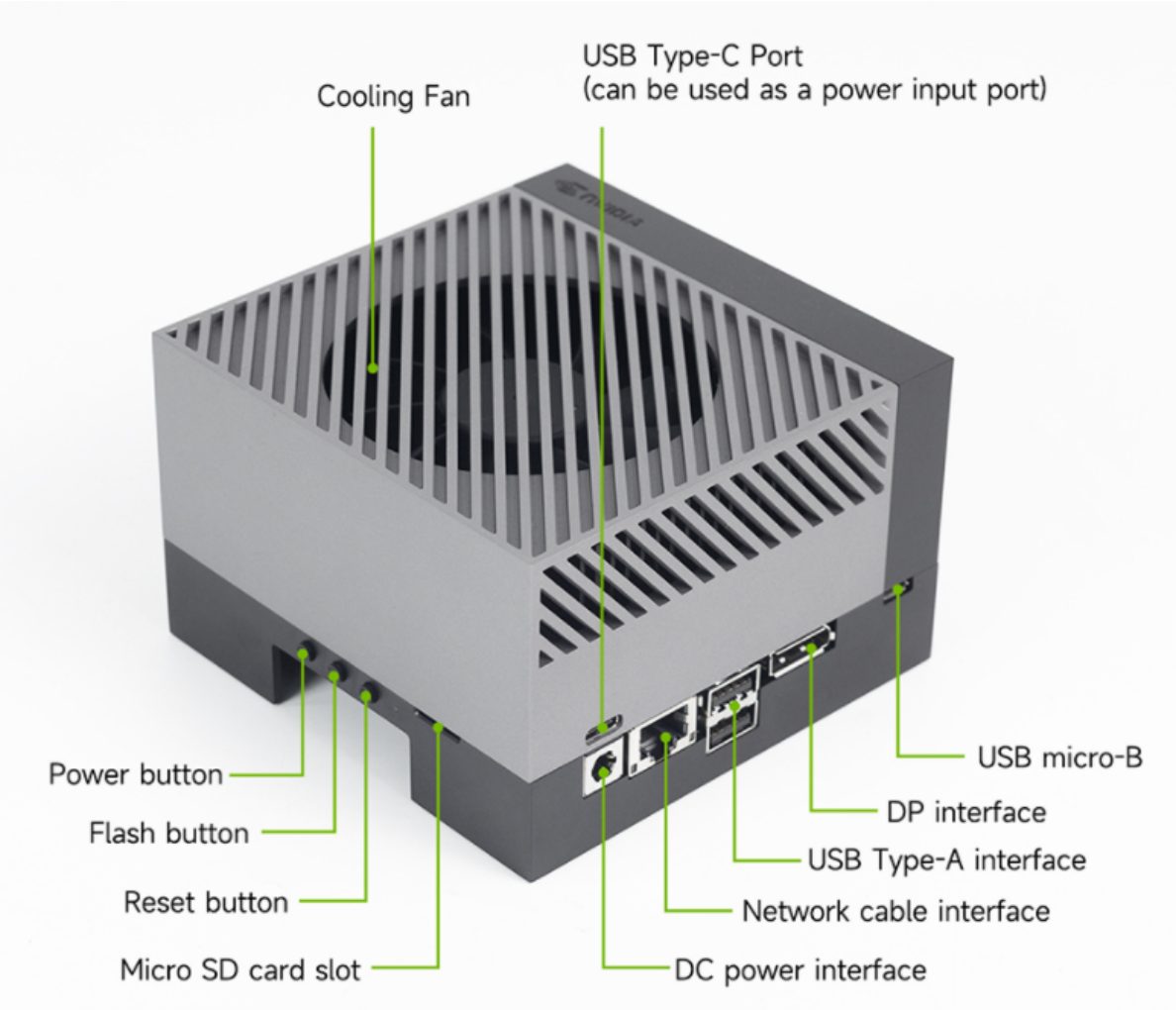
First, follow the steps in [**Chapter 2 Jetson basic course: 02.Flash new system to the onboard eMMC**] to burn the system. Then, power off the computer. Next, install the factory-installed SSD onto the Jetson AGX Orin motherboard. Connect the DP cable (to the monitor) and peripherals such as the mouse and keyboard. Use the included power adapter to boot the motherboard system.

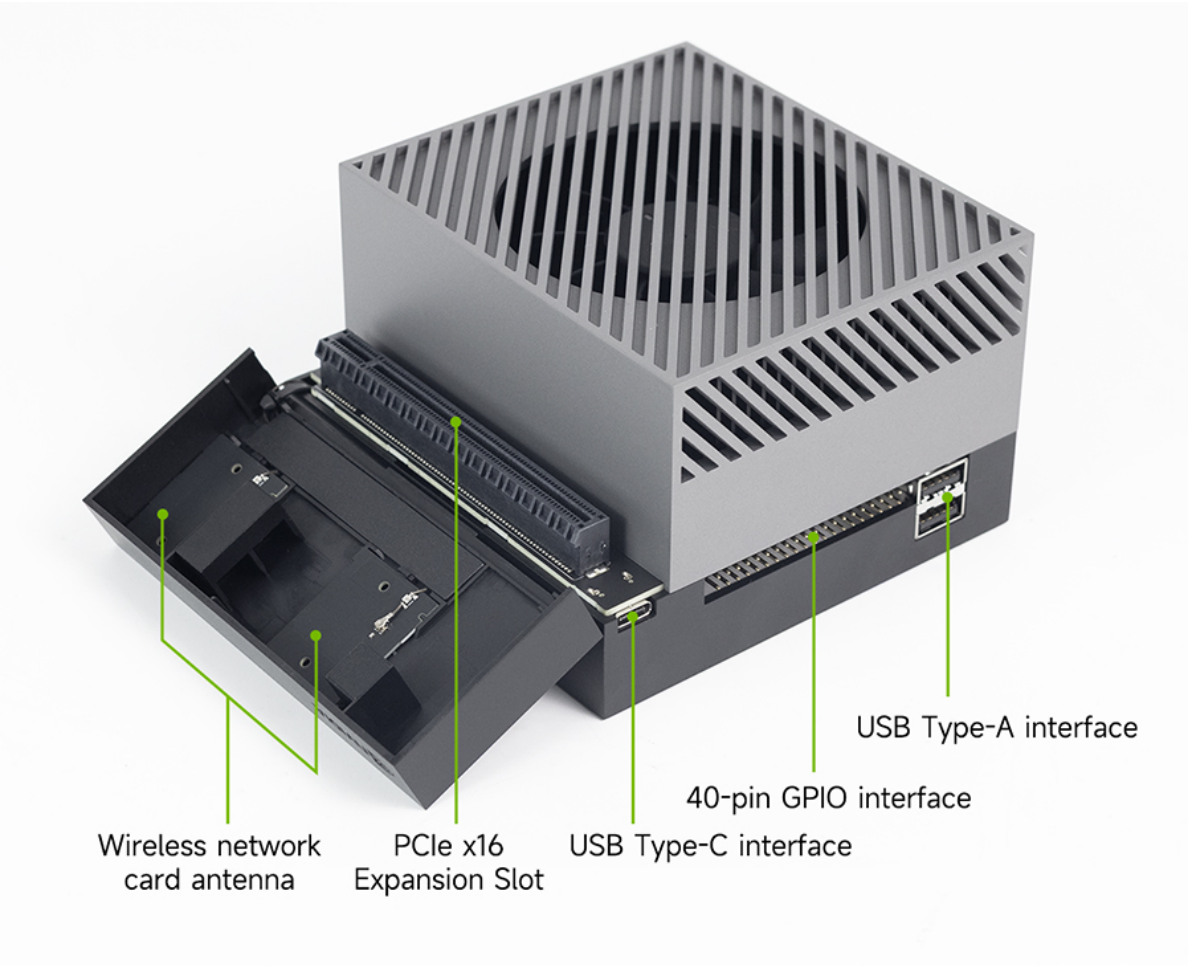
The power adapter included with the Jetson AGX Orin official kit has a Type-C interface. Both Type-C ports on the motherboard can be used as power input interfaces. For convenience, generally, the power adapter should be plugged into the Type-C port above the DC power connector. The image below shows a power connection diagram.

By default, the AUTOMATION header on the bottom of the motherboard will appear as shown in the image below. Plug in the power, and the computer will automatically boot into the system.



For other interface connections, please connect according to your needs and the motherboard interface instructions.


Cooling Fan
USB Type-C Port
(can be used as a power input port)
Power button
Flash button
Reset button
Micro SD card slot
USB micro-B
DP interface
USB Type-A interface
Network cable interface
DC power interface


USB Type-A interface
40-pin GPIO interface
Wireless network
card antenna
PCIe x16
Expansion Slot
USB Type-C interface

Jetson AGX Orin series motherboards require a monitor to display the system desktop correctly. For first-time users, it is recommended to connect an external monitor and configure VNC remote login.

Note: According to user feedback, some users can use a graphics card spoofing tool and virtual desktop to achieve VNC remote login. This is not within the scope of this tutorial!

2. System Environment

2.1. Basic Information

System Username: jetson

System Password: yahboom

Root Password: yahboom

2.2. Jupyter Lab

Host Access: http://localhost:8888/

LAN Access: http://motherboard IP:8888/

Access Password: yahboom

2.3. Label Studio (Requires manual startup)

Host Access: http://localhost:8081/

LAN Access: http://motherboard IP:8081/

Access Username: yahboom@163.com

Access Password: yahboom@163.com

2.4. Accessing Docker

```
docker run -it ubuntu:18.04 /bin/bash
```

2.5. Open WebUI

Host Access: http://localhost:8080/

LAN Access: http://motherboard IP:8080/

Username: yahboom

Email: yahboom@163.com

Password: yahboom

3. Notes

- Due to software version iterations and different user environments, we cannot guarantee that everyone can successfully set up the environment. The only thing we can provide is a complete system image. If users encounter environment problems that they cannot solve themselves, please use the factory image we provide for tutorial learning.